STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 10: Thermal Properties of Matter

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

What is temperature’s role at equilibrium? [1]
Convert 27 deg C to kelvin. [1]
Specific heat capacity? [1]
Heat for temperature change without phase change? [1]
Latent heat relation? [1]
Heat transfer through vacuum? [1]
Ideal gas equation? [1]
Newton’s law of cooling? [1]

2 kg warmed 5 K, c=400: Q? [2]
=1.2x10^-5,L=2m,DeltaT=50K: extension? [2]
100 deg C in kelvin? [2]
Heat capacity of .5 kg with c=800? [2]
Heat to melt .2 kg ice, L=3.34x10^5? [2]
Why is temperature constant during boiling? [2]

.5 kg water c=4200 heated 10 deg C: heat? [3]
Rod L=2m,=1.5x10^-5,DeltaT=40K: expansion? [3]
Mix .2 kg at 80 deg C and .3 kg at 20 deg C, equal c, no heat loss: final T? [3]
Gas at fixed volume warms 300K to 450K; initial p=100kPa. Final p? [3]

Describe three heat-transfer modes. [4]
Melt .2 kg ice at 0 deg C then warm water by 20K; L=3.34x10^5,c=4200. Total Q? [4]

End of paper - check working, units, signs and significant figures.